LABORATOR 2

CORCODEL ANA-MARIA

GRUPA 10 216

ACTIVITATEA 1:

```
vboxuser@Ubuntu: ~/Desktop
  GNU nano 7.2                    ex3_lab2.c
#include <omp.h>
#include <stdio.h>
#include <stdlib.h>

#define CHUNKSIZE 10
#define N 100000

int main() {
    int nthreads, tid, i, chunk;
    float a[N], b[N], c[N];

    for (i = 0; i < N; i++) {
        a[i] = i * 1.0f;
        b[i] = i * 1.0f;
        c[i] = 0.0f;
    }

    chunk = CHUNKSIZE;

    double t1, t2;
                          [ Read 55 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

vboxuser@Ubuntu: ~/Desktop

GNU nano 7.2 ex3_lab2.c *

```
    chunk = CHUNKSIZE;

    double t1, t2;
    t1 = omp_get_wtime();

    #pragma omp parallel shared(a, b, c, nthreads, chunk) private(i, tid)
    {
        tid = omp_get_thread_num();

        if (tid == 0) {
            nthreads = omp_get_num_threads();
            printf("Number of threads = %d\n", nthreads);
        }

        /*
           Pentru teste, schimbăm doar linia de mai jos între:

           #pragma omp for schedule(static)
           #pragma omp for schedule(static,1)
           #pragma omp for schedule(static,2)
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

vboxuser@Ubuntu: ~/Desktop

GNU nano 7.2 ex3_lab2.c *

```
           #pragma omp for schedule(static,2)
           #pragma omp for schedule(dynamic,1)
           #pragma omp for schedule(dynamic,2)
           #pragma omp for schedule(guided,1)
           #pragma omp for schedule(guided,2)
           #pragma omp for schedule(auto)
        */

        #pragma omp for schedule(auto)
        for (i = 0; i < N; i++) {
            c[i] = a[i] + b[i];
        }   }

    t2 = omp_get_wtime();

    printf("Execution time auto: %g\n", t2 - t1);

    return 0;
}
```

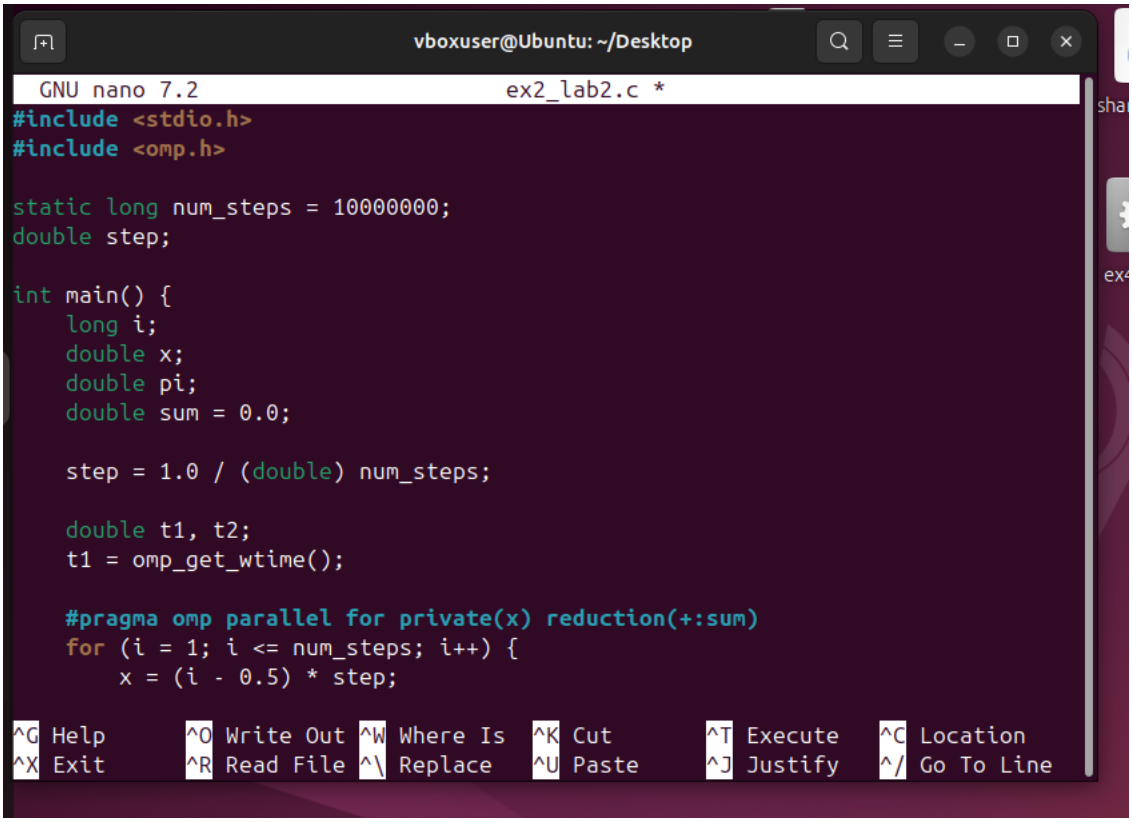
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

```
vboxuser@Ubuntu:~/Desktop$ nano ex2_lab2.c
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time: 0.00707992
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time: 0.00939979
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time static,1: 0.0133998
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time static,2: 0.00567025
vboxuser@Ubuntu:~/Desktop$ S
```

```
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time dynamic,1: 0.0128059
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time dynamic,2: 0.0123264
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time guided,1: 0.00721868
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time guided,2: 0.00884127
vboxuser@Ubuntu:~/Desktop$ nano ex3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex3_lab2.c -o ex3_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex3_lab2
Number of threads = 8
Execution time auto: 0.00384535
vboxuser@Ubuntu:~/Desktop$
```

Tip Schedule	Chunk Size	Timp de execuție
dynamic	default	0.000833
auto	auto	0.000649
guided	1	0.000539
guided	2	0.000989
static	default	0.001232
static	1	0.000870
static	2	0.000934
static	4	0.000681

ACTIVITATEA 2:



```
vboxuser@Ubuntu: ~/Desktop
GNU nano 7.2                    ex2_lab2.c *

    double t1, t2;
    t1 = omp_get_wtime();

    #pragma omp parallel for private(x) reduction(+:sum)
    for (i = 1; i <= num_steps; i++) {
        x = (i - 0.5) * step;
        sum = sum + 4.0 / (1.0 + x * x);
    }

    t2 = omp_get_wtime();

    pi = step * sum;

    printf("Execution time: %g\n", t2 - t1);
    printf("The computed value of pi is %f\n", pi);

    return 0;
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
vboxuser@Ubuntu:~/Desktop$ nano ex2_lab2.c
vboxuser@Ubuntu:~/Desktop$ nano ex2_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp ex2_lab2.c -o ex2_lab2
vboxuser@Ubuntu:~/Desktop$ ./ex2_lab2
Execution time: 0.0323323
The computed value of pi is 3.141593
vboxuser@Ubuntu:~/Desktop$ ./ex2_lab2
Execution time: 0.0416045
The computed value of pi is 3.141593
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=1
vboxuser@Ubuntu:~/Desktop$ ./ex2_lab2
Execution time: 0.0393129
The computed value of pi is 3.141593
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=4
vboxuser@Ubuntu:~/Desktop$ ./ex2_lab2
Execution time: 0.0357508
The computed value of pi is 3.141593
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=8
vboxuser@Ubuntu:~/Desktop$ ./ex2_lab2
Execution time: 0.017908
The computed value of pi is 3.141593
vboxuser@Ubuntu:~/Desktop$
```

ACTIVITATEA 3:

GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>

#define N 24

static void do_work(int i) {
    volatile double x = 0.0;

    /* workload neuniform: unele iterații sunt mai grele decât altele */
    int work = (i % 7 + 1) * 2000000;

    for (int k = 0; k < work; k++) {
        x += 0.0000001 * k;
    }
}

int main() {
    int owner[N];
    for (int i = 0; i < N; i++) {
        owner[i] = -1;
    }

    double t1 = omp_get_wtime();

    #pragma omp parallel
    {
        int tid = omp_get_thread_num();

        /*
           SCHIMBĂ DOAR LINIA ASTA între variantele cerute:

           #pragma omp for schedule(dynamic,2)
           #pragma omp for schedule(guided,2)
```

GNU nano 7.2

```
    }

    double t1 = omp_get_wtime();

    #pragma omp parallel
    {
        int tid = omp_get_thread_num();

        /*
           SCHIMBĂ DOAR LINIA ASTA intre variantele cerute:

           #pragma omp for schedule(dynamic,2)
           #pragma omp for schedule(guided,2)
        */
        #pragma omp for schedule(guided,2)
        for (int i = 0; i < N; i++) {
            owner[i] = tid;
            do_work(i);
            printf("Thread %d a executat iteratia %d\n", tid, i);
        }
    }

    double t2 = omp_get_wtime();

    printf("\nRezumat distributie iteratii:\n");
    for (int i = 0; i < N; i++) {
        printf("iteratia %2d -> thread %d\n", i, owner[i]);
    }

    printf("\nTimp total: %f secunde\n", t2 - t1);

    return 0;
}
```

DINAMIC:

```
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=4
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 1 a executat iteratia 0
Thread 3 a executat iteratia 2
Thread 1 a executat iteratia 1
Thread 2 a executat iteratia 4
Thread 1 a executat iteratia 8
Thread 3 a executat iteratia 3
Thread 0 a executat iteratia 6
Thread 0 a executat iteratia 7
Thread 1 a executat iteratia 9
Thread 1 a executat iteratia 14
Thread 3 a executat iteratia 10
Thread 1 a executat iteratia 15
Thread 2 a executat iteratia 5
Thread 0 a executat iteratia 12
Thread 1 a executat iteratia 16
Thread 3 a executat iteratia 11
Thread 2 a executat iteratia 18
Thread 1 a executat iteratia 17
Thread 1 a executat iteratia 22
Thread 0 a executat iteratia 13
Thread 3 a executat iteratia 20
Thread 1 a executat iteratia 23
Thread 2 a executat iteratia 19
Thread 3 a executat iteratia 21
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 1
iteratia  1 -> thread 1
iteratia  2 -> thread 3
iteratia  3 -> thread 3
iteratia  4 -> thread 2
iteratia  5 -> thread 2
iteratia  6 -> thread 0
iteratia  7 -> thread 0
iteratia  8 -> thread 1
iteratia  9 -> thread 1
iteratia 10 -> thread 3
iteratia 11 -> thread 3
iteratia 12 -> thread 0
iteratia 13 -> thread 0
iteratia 14 -> thread 1
iteratia 15 -> thread 1
iteratia 16 -> thread 1
iteratia 17 -> thread 1
iteratia 18 -> thread 2
iteratia 19 -> thread 2
iteratia 20 -> thread 3
iteratia 21 -> thread 3
iteratia 22 -> thread 1
iteratia 23 -> thread 1

Timp total: 0.179813 secunde
```

```
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 0 a executat iteratia 0
Thread 0 a executat iteratia 1
Thread 2 a executat iteratia 2
Thread 0 a executat iteratia 8
Thread 1 a executat iteratia 4
Thread 3 a executat iteratia 6
Thread 2 a executat iteratia 3
Thread 0 a executat iteratia 9
Thread 3 a executat iteratia 7
Thread 3 a executat iteratia 14
Thread 1 a executat iteratia 5
Thread 3 a executat iteratia 15
Thread 2 a executat iteratia 10
Thread 1 a executat iteratia 18
Thread 3 a executat iteratia 16
Thread 0 a executat iteratia 12
Thread 1 a executat iteratia 19
Thread 3 a executat iteratia 17
Thread 0 a executat iteratia 13
Thread 3 a executat iteratia 22
Thread 2 a executat iteratia 11
Thread 3 a executat iteratia 23
Thread 1 a executat iteratia 20
Thread 1 a executat iteratia 21
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 0
iteratia  1 -> thread 0
iteratia  2 -> thread 2
iteratia  3 -> thread 2
iteratia  4 -> thread 1
iteratia  5 -> thread 1
iteratia  6 -> thread 3
iteratia  7 -> thread 3
iteratia  8 -> thread 0
iteratia  9 -> thread 0
iteratia 10 -> thread 2
iteratia 11 -> thread 2
iteratia 12 -> thread 0
iteratia 13 -> thread 0
iteratia 14 -> thread 3
iteratia 15 -> thread 3
iteratia 16 -> thread 3
iteratia 17 -> thread 3
iteratia 18 -> thread 1
iteratia 19 -> thread 1
iteratia 20 -> thread 1
iteratia 21 -> thread 1
iteratia 22 -> thread 3
iteratia 23 -> thread 3

Timp total: 0.248494 secunde
```

```
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 3 a executat iteratia 0
Thread 1 a executat iteratia 2
Thread 3 a executat iteratia 1
Thread 3 a executat iteratia 8
Thread 1 a executat iteratia 3
Thread 0 a executat iteratia 6
Thread 2 a executat iteratia 4
Thread 0 a executat iteratia 7
Thread 3 a executat iteratia 9
Thread 3 a executat iteratia 14
Thread 3 a executat iteratia 15
Thread 0 a executat iteratia 12
Thread 3 a executat iteratia 16
Thread 1 a executat iteratia 10
Thread 3 a executat iteratia 17
Thread 2 a executat iteratia 5
Thread 0 a executat iteratia 13
Thread 3 a executat iteratia 18
Thread 0 a executat iteratia 22
Thread 1 a executat iteratia 11
Thread 3 a executat iteratia 19
Thread 0 a executat iteratia 23
Thread 2 a executat iteratia 20
Thread 2 a executat iteratia 21
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 3
iteratia  1 -> thread 3
iteratia  2 -> thread 1
iteratia  3 -> thread 1
iteratia  4 -> thread 2
iteratia  5 -> thread 2
iteratia  6 -> thread 0
iteratia  7 -> thread 0
iteratia  8 -> thread 3
iteratia  9 -> thread 3
iteratia 10 -> thread 1
iteratia 11 -> thread 1
iteratia 12 -> thread 0
iteratia 13 -> thread 0
iteratia 14 -> thread 3
iteratia 15 -> thread 3
iteratia 16 -> thread 3
iteratia 17 -> thread 3
iteratia 18 -> thread 3
iteratia 19 -> thread 3
iteratia 20 -> thread 2
iteratia 21 -> thread 2
iteratia 22 -> thread 0
iteratia 23 -> thread 0

Timp total: 0.321824 secunde
```

GUIDE:

```
vboxuser@Ubuntu:~/Desktop$ nano activitatea3_lab2.c
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=4
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 3 a executat iteratia 0
Thread 3 a executat iteratia 1
Thread 2 a executat iteratia 2
Thread 3 a executat iteratia 8
Thread 0 a executat iteratia 4
Thread 3 a executat iteratia 9
Thread 2 a executat iteratia 3
Thread 1 a executat iteratia 6
Thread 1 a executat iteratia 7
Thread 1 a executat iteratia 14
Thread 3 a executat iteratia 10
Thread 1 a executat iteratia 15
Thread 1 a executat iteratia 16
Thread 3 a executat iteratia 11
Thread 0 a executat iteratia 5
Thread 1 a executat iteratia 17
Thread 3 a executat iteratia 18
Thread 2 a executat iteratia 12
Thread 1 a executat iteratia 22
Thread 3 a executat iteratia 19
Thread 1 a executat iteratia 23
Thread 0 a executat iteratia 20
Thread 2 a executat iteratia 13
Thread 0 a executat iteratia 21
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 3
iteratia  1 -> thread 3
iteratia  2 -> thread 2
iteratia  3 -> thread 2
iteratia  4 -> thread 0
iteratia  5 -> thread 0
iteratia  6 -> thread 1
iteratia  7 -> thread 1
iteratia  8 -> thread 3
iteratia  9 -> thread 3
iteratia 10 -> thread 3
iteratia 11 -> thread 3
iteratia 12 -> thread 2
iteratia 13 -> thread 2
iteratia 14 -> thread 1
iteratia 15 -> thread 1
iteratia 16 -> thread 1
iteratia 17 -> thread 1
iteratia 18 -> thread 3
iteratia 19 -> thread 3
iteratia 20 -> thread 0
iteratia 21 -> thread 0
iteratia 22 -> thread 1
iteratia 23 -> thread 1

Timp total: 0.301279 secunde
```

```
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 1 a executat iteratia 0
Thread 1 a executat iteratia 1
Thread 3 a executat iteratia 2
Thread 1 a executat iteratia 8
Thread 3 a executat iteratia 3
Thread 1 a executat iteratia 9
Thread 0 a executat iteratia 4
Thread 3 a executat iteratia 10
Thread 1 a executat iteratia 12
Thread 2 a executat iteratia 6
Thread 3 a executat iteratia 11
Thread 0 a executat iteratia 5
Thread 2 a executat iteratia 7
Thread 3 a executat iteratia 14
Thread 3 a executat iteratia 15
Thread 0 a executat iteratia 16
Thread 1 a executat iteratia 13
Thread 1 a executat iteratia 22
Thread 2 a executat iteratia 18
Thread 0 a executat iteratia 17
Thread 1 a executat iteratia 23
Thread 3 a executat iteratia 20
Thread 2 a executat iteratia 19
Thread 3 a executat iteratia 21
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 1
iteratia  1 -> thread 1
iteratia  2 -> thread 3
iteratia  3 -> thread 3
iteratia  4 -> thread 0
iteratia  5 -> thread 0
iteratia  6 -> thread 2
iteratia  7 -> thread 2
iteratia  8 -> thread 1
iteratia  9 -> thread 1
iteratia 10 -> thread 3
iteratia 11 -> thread 3
iteratia 12 -> thread 1
iteratia 13 -> thread 1
iteratia 14 -> thread 3
iteratia 15 -> thread 3
iteratia 16 -> thread 0
iteratia 17 -> thread 0
iteratia 18 -> thread 2
iteratia 19 -> thread 2
iteratia 20 -> thread 3
iteratia 21 -> thread 3
iteratia 22 -> thread 1
iteratia 23 -> thread 1

Timp total: 0.307744 secunde
```

```
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 0 a executat iteratia 0
Thread 0 a executat iteratia 1
Thread 3 a executat iteratia 2
Thread 0 a executat iteratia 8
Thread 1 a executat iteratia 4
Thread 3 a executat iteratia 3
Thread 0 a executat iteratia 9
Thread 1 a executat iteratia 5
Thread 3 a executat iteratia 10
Thread 1 a executat iteratia 14
Thread 0 a executat iteratia 12
Thread 1 a executat iteratia 15
Thread 2 a executat iteratia 6
Thread 2 a executat iteratia 7
Thread 3 a executat iteratia 11
Thread 1 a executat iteratia 16
Thread 0 a executat iteratia 13
Thread 3 a executat iteratia 20
Thread 3 a executat iteratia 21
Thread 0 a executat iteratia 22
Thread 1 a executat iteratia 17
Thread 0 a executat iteratia 23
Thread 2 a executat iteratia 18
Thread 2 a executat iteratia 19

Rezumat distributie iteratii:
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 0
iteratia  1 -> thread 0
iteratia  2 -> thread 3
iteratia  3 -> thread 3
iteratia  4 -> thread 1
iteratia  5 -> thread 1
iteratia  6 -> thread 2
iteratia  7 -> thread 2
iteratia  8 -> thread 0
iteratia  9 -> thread 0
iteratia 10 -> thread 3
iteratia 11 -> thread 3
iteratia 12 -> thread 0
iteratia 13 -> thread 0
iteratia 14 -> thread 1
iteratia 15 -> thread 1
iteratia 16 -> thread 1
iteratia 17 -> thread 1
iteratia 18 -> thread 2
iteratia 19 -> thread 2
iteratia 20 -> thread 3
iteratia 21 -> thread 3
iteratia 22 -> thread 0
iteratia 23 -> thread 0

Timp total: 0.311013 secunde
```

STATIC:

```
vboxuser@Ubuntu:~/Desktop$ nano activitatea3_lab2.c
vboxuser@Ubuntu:~/Desktop$ nano activitatea3_lab2.c
vboxuser@Ubuntu:~/Desktop$ gcc -fopenmp activitatea3_lab2.c -o activitatea3_lab2
vboxuser@Ubuntu:~/Desktop$ ./activitatea3_lab2
Thread 0 a executat iteratia 0
Thread 1 a executat iteratia 1
Thread 2 a executat iteratia 2
Thread 3 a executat iteratia 3
Thread 3 a executat iteratia 7
Thread 0 a executat iteratia 4
Thread 1 a executat iteratia 5
Thread 3 a executat iteratia 11
Thread 3 a executat iteratia 15
Thread 0 a executat iteratia 8
Thread 2 a executat iteratia 6
Thread 1 a executat iteratia 9
Thread 3 a executat iteratia 19
Thread 3 a executat iteratia 23
Thread 2 a executat iteratia 10
Thread 2 a executat iteratia 14
Thread 0 a executat iteratia 12
Thread 1 a executat iteratia 13
Thread 0 a executat iteratia 16
Thread 1 a executat iteratia 17
Thread 1 a executat iteratia 21
Thread 2 a executat iteratia 18
Thread 2 a executat iteratia 22
Thread 0 a executat iteratia 20
```

```
Rezumat distributie iteratii:
iteratia  0 -> thread 0
iteratia  1 -> thread 1
iteratia  2 -> thread 2
iteratia  3 -> thread 3
iteratia  4 -> thread 0
iteratia  5 -> thread 1
iteratia  6 -> thread 2
iteratia  7 -> thread 3
iteratia  8 -> thread 0
iteratia  9 -> thread 1
iteratia 10 -> thread 2
iteratia 11 -> thread 3
iteratia 12 -> thread 0
iteratia 13 -> thread 1
iteratia 14 -> thread 2
iteratia 15 -> thread 3
iteratia 16 -> thread 0
iteratia 17 -> thread 1
iteratia 18 -> thread 2
iteratia 19 -> thread 3
iteratia 20 -> thread 0
iteratia 21 -> thread 1
iteratia 22 -> thread 2
iteratia 23 -> thread 3

Timp total: 0.271733 secunde
```

ACTIVITATEA 4:

GNU nano 7.2 activitatea4_lab2.c

```
#include <omp.h>
#include <stdio.h>
#include <stdlib.h>
#include <string>
#include <vector>
#include <iostream>

using namespace std;

int main() {
    vector<string> filenames;

    for (int i = 1; i <= 100; i++) {
        filenames.push_back("files/f" + to_string(i));
    }

    int global_letters[128] = {0};

    double t1 = omp_get_wtime();

    #pragma omp parallel
    {
        int local_letters[128] = {0};

        #pragma omp for schedule(dynamic,1)
        for (int i = 0; i < (int)filenames.size(); i++) {
            FILE *file = fopen(filenames[i].c_str(), "r");
            if (file == NULL) {
                continue;
            }

            char content[10000];

            while (fgets(content, sizeof(content), file) != NULL) {
```

GNU nano 7.2 activitatea4_lab2.c

```
            char content[10000];

            while (fgets(content, sizeof(content), file) != NULL) {
                for (int j = 0; content[j] != '\0'; j++) {
                    unsigned char ch = (unsigned char)content[j];
                    if (ch >= 32 && ch < 128) {
                        local_letters[ch]++;
                    }
                }
            }

            fclose(file);
        }

        #pragma omp critical
        {
            for (int i = 32; i < 128; i++) {
                global_letters[i] += local_letters[i];
            }
        }
    }

    double t2 = omp_get_wtime();

    for (int i = 32; i < 128; i++) {
        cout << (char)i << ": " << global_letters[i] << endl;
    }

    cout << "Execution time: " << (t2 - t1) << endl;

    return 0;
}
```

```
vboxuser@Ubuntu:~/Desktop$ g++ activitatea4_lab2.c -O2 -fopenmp -o activitatea4_lab2
vboxuser@Ubuntu:~/Desktop$ chmod +x gen_files.sh
chmod: cannot access 'gen_files.sh': No such file or directory
vboxuser@Ubuntu:~/Desktop$ nano gen_files.sh
vboxuser@Ubuntu:~/Desktop$ chmod +x gen_files.sh
vboxuser@Ubuntu:~/Desktop$ ./gen_files.sh
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=4
vboxuser@Ubuntu:~/Desktop$ ./activitatea4_lab2
 : 0
!: 0
": 0
#: 0
$: 0
%: 0
&: 0
': 0
(: 0
): 0
*: 0
+: 25728
,: 0
-: 0
.: 0
/: 25605
0: 25731
1: 25555
2: 25639
3: 25631
4: 25524
5: 25743
6: 25568
7: 25593
8: 25554
9: 25518
:: 0
```

```
8: 25554
9: 25518
:: 0
;: 0
<: 0
=: 0
>: 0
?: 0
@: 0
A: 25582
B: 25894
C: 25800
D: 25593
E: 25676
F: 25549
G: 25687
H: 25441
I: 25410
J: 25475
K: 25743
L: 25535
M: 25604
N: 25669
O: 25731
P: 25575
Q: 25348
R: 25554
S: 25515
T: 25483
U: 25448
V: 25647
W: 25447
X: 25799
Y: 25807
Z: 25526
[: 0
\: 0
]: 0
^: 0
```

```
\: 0
]: 0
^: 0
_: 0
`: 0
a: 25883
b: 25778
c: 25519
d: 25489
e: 25095
f: 25385
g: 25731
h: 25553
i: 25601
j: 25345
k: 25705
l: 25628
m: 25534
n: 25399
o: 25819
p: 25375
q: 25519
r: 25933
s: 25465
t: 25601
u: 25619
v: 25795
w: 25545
x: 25434
y: 25313
z: 25481
{: 0
|: 0
}: 0
~: 0
: 0
Execution time: 0.0518142
vboxuser@Ubuntu:~/Desktop$
```

Static:

```
vboxuser@Ubuntu:~/Desktop$ nano activitatea4_lab2.c
vboxuser@Ubuntu:~/Desktop$ nano activitatea4_lab2.c
vboxuser@Ubuntu:~/Desktop$ g++ activitatea4_lab2.c -O2 -fopenmp -o activitatea4_lab2
vboxuser@Ubuntu:~/Desktop$ chmod +x gen_files.sh
vboxuser@Ubuntu:~/Desktop$ ./gen_files.sh
vboxuser@Ubuntu:~/Desktop$ export OMP_NUM_THREADS=4
vboxuser@Ubuntu:~/Desktop$ ./activitatea4_lab2
 : 0
!: 0
": 0
#: 0
$: 0
%: 0
&: 0
': 0
(: 0
): 0
*: 0
+: 26144
,: 0
-: 0
.: 0
/: 26054
0: 26525
1: 26350
2: 26379
3: 26516
4: 26218
5: 26310
6: 26044
7: 26168
8: 26265
9: 26262
:: 0
;: 0
<: 0
=: 0
>: 0
```

```
^: 0
_: 0
`: 0
a: 26209
b: 26252
c: 26421
d: 26342
e: 26186
f: 26190
g: 26253
h: 26277
i: 25887
j: 26184
k: 26347
l: 26427
m: 26198
n: 26170
o: 26209
p: 26086
q: 26522
r: 26427
s: 26219
t: 26326
u: 26157
v: 26458
w: 26276
x: 26597
y: 26350
z: 26396
{: 0
|: 0
}: 0
~: 0
: 0
Execution time: 0.14984
vboxuser@Ubuntu:~/Desktop$ ./activitatea4_lab2
```

```
vboxuser@Ubuntu:~/Desktop$ ./activitatea4_lab2
 : 0
!: 0
": 0
#: 0
$: 0
%: 0
&: 0
': 0
(: 0
): 0
*: 0
+: 26144
,: 0
-: 0
.: 0
/: 26054
0: 26525
1: 26350
2: 26379
3: 26516
4: 26218
5: 26310
6: 26044
7: 26168
8: 26265
9: 26262
:: 0
;: 0
<: 0
=: 0
>: 0
?: 0
@: 0
A: 26174
B: 26461
```

```
\: 0
]: 0
^: 0
_: 0
`: 0
a: 26209
b: 26252
c: 26421
d: 26342
e: 26186
f: 26190
g: 26253
h: 26277
i: 25887
j: 26184
k: 26347
l: 26427
m: 26198
n: 26170
o: 26209
p: 26086
q: 26522
r: 26427
s: 26219
t: 26326
u: 26157
v: 26458
w: 26276
x: 26597
y: 26350
z: 26396
{: 0
|: 0
}: 0
~: 0
: 0
Execution time: 0.042891
vboxuser@Ubuntu:~/Desktop$ S
```